

Abstract.

1 Introduction

1.1 Mechanical models of snow stability and their input parameters

1.2 Snowpits and the SnowPilot database

5 A snowpit is a trench excavated into the snowpack, typically to the ground or to a depth of interest. For each stratigraphic layer encountered, the observer records hand hardness, grain form, grain size, and layer thickness; density may also be measured directly on individual layers using a density cutter and scale. Terrain characteristics (slope angle, aspect, elevation) and weather conditions at the time of the observation supplement the stratigraphy. Stability tests are commonly conducted alongside the snow profile – most frequently the Compression Test (CT), Extended Column Test (ECT), and Propagation Saw Test (PST) –
10 and record the stress required to initiate a fracture, whether that fracture propagates, and the depth and character of the failure plane. Standard observation protocols are described in ?, ?, ?, and ?.

SnowPilot (??) is a free, web-based application for recording, visualising, and storing snowpit observations. Data are archived in the Canadian Avalanche Association Markup Language (CAAML) XML format (?), an open community standard for snow and avalanche data exchange. The database is community-sourced, with contributors spanning avalanche centre
15 forecasters, ski patrol, field researchers, and backcountry educators, and is particularly widely used in North America.

A summary of the SnowPilot data used in this analysis – the decade from 1 September 2014 to 31 August 2025 – is provided in Table 1.

The scale of the SnowPilot dataset – 51,138 snowpits over ten years, spanning 35 countries and 5,440 unique observers – makes it uniquely suited for population-level quantification of observational coverage and parameterization sensitivity. Critically, because the data are community-sourced rather than collected under controlled research conditions, coverage fractions
20 derived from this dataset reflect the realistic distribution of field observation completeness encountered by practitioners, rather than the more complete data typical of targeted research campaigns.

Table 1. Summary of the SnowPilot database for the analysis period 1 September 2014 to 31 August 2025.

	Count
<i>Snowpits</i>	
Total	51,138 ^a
Unique observers	5,440
Countries represented	35
With slope angle observation	46,226 (90.4 %)
Recorded near an avalanche	1,586 (3.1 %)
<i>Snow layers</i>	
Total	377,729
With hand hardness	342,690 (90.7 %)
With primary grain form	309,232 (81.9 %)
With grain size	179,342 (47.5 %)
With direct density measurement ^b	10,692 (2.8 %)
<i>Stability tests</i>	
Compression tests (CT)	52,305
Extended column tests (ECT)	48,391
Propagation saw tests (PST)	6,326

^a One file (snowpits-55240-caaml.xml) failed to parse and was excluded, leaving 51,138 snowpits.

^b SnowPilot allows density to be entered either as an attribute of a stratigraphic layer or as a separate density profile. Our analysis uses only density observations whose depth and thickness match a stratigraphic layer boundary; the count reported here reflects this aligned subset.

1.3 Objectives and scope

2 Data and methods

25 2.1 Data: SnowPilot and SnowPylot

SnowPylot (?) is an open-source Python library for parsing SnowPilot CAAML XML files into structured Python objects. It provides the data ingestion layer for this analysis, translating the raw CAAML archive into an accessible object hierarchy from which measured field observations are extracted for parameterization.

Each CAAML file is parsed into a `SnowPit` object with two main sub-objects. The `snow_profile` object contains the ordered list of stratigraphic `Layer` objects — each carrying hand hardness, primary grain form, grain size, and layer thickness — alongside a separate `density_profile` list of density observations, each recording a depth, thickness, and density value. The `stability_tests` object contains result lists for each test type; the `ECT` list is the relevant one for this analysis, with each entry recording the failure depth, number of taps, and propagation result.

Three data handling decisions govern which observations are included in the analysis.

35 *Density alignment.* SnowPilot allows density to be entered either as an attribute of an individual stratigraphic layer or as a separate density profile. When entered as a separate profile, a density observation cannot always be unambiguously matched to a stratigraphic layer boundary because the recorded depth and thickness intervals do not always coincide with layer boundaries. This analysis uses only density observations whose recorded depth and thickness match a layer boundary exactly; density profile entries that cannot be matched are excluded. Of the 19,877 density observations in the database, 10,692 (2.8 % of all

40 layers) could be unambiguously aligned and are carried forward as measured density on the corresponding layer (see also Table 1, footnote b).

Grain form resolution. SnowPilot records grain forms using CAAML grain form codes, which may be either two-character basic grain class codes (e.g., RG for rounded grains, FC for faceted crystals) or longer sub-grain class codes (e.g., RGmx for mixed rounded grains, FCxr for depth-hoar-like facets). Each parameterization method defines a set of valid grain form codes.

45 When a layer’s recorded grain form code is not directly valid for a given method, the code is truncated to its two-character basic grain class prefix and re-evaluated; if the basic class is valid for the method, it is used. If neither the full code nor the truncated prefix is valid for the method, the method is inapplicable to that layer and the pathway fails silently for it.

Primary grain form. Each layer in SnowPilot may carry both a primary and a secondary grain form. This analysis uses only the primary grain form for all parameterization calculations; secondary grain form information is not used.

50 2.2 Measurement uncertainties

2.3 Parameterizations for mechanical properties

2.3.1 Density

2.3.2 Effective, isotropic Young’s modulus

2.3.3 Effective, isotropic Poisson’s ratio

55 2.3.4 Shear modulus

2.3.5 Plate theory parameters for composite slabs

2.4 Parameterization graph and calculation pathways

The parameterizations described in Sect. 2.3 are implemented within a directed acyclic graph (DAG) that encodes all valid calculation pathways from snowpit observations to target mechanical parameters. This section describes the structure of the graph, the algorithm for enumerating pathways, and the execution engine that applies each pathway to a slab.

2.4.1 Representing parameterizations as a directed graph

The parameterization graph contains two types of nodes. *Parameter nodes* represent quantities — either directly observed field measurements (hand hardness, grain form, grain size, directly measured density, and layer thickness) or derived parameters (density, Young’s modulus, Poisson’s ratio, shear modulus, and the plate-theory stiffness parameters A11, B11, D11, and A55).
65 A parameter node implements OR logic: any single incoming method edge is sufficient to determine the parameter, so the graph represents all alternative estimation methods for a given quantity without requiring any one of them. *Merge nodes* implement AND logic: each merge node represents a specific combination of inputs required by one or more calculation methods, and all inputs feeding a merge node must be available before the corresponding method can be applied. Each merge node has exactly one outgoing edge to its output parameter node.

70 Edges are directed and of two kinds. Data-flow edges carry observed or previously computed values forward from a parameter node to a merge node. Method edges carry the result of a calculation from a merge node to the output parameter node, and are labelled with the method name (for example, *geldsetzer*, *wautier*, or *kochle*). The root node, representing the snowpit observation, connects by data-flow edges to all directly measured quantities.

A key structural feature of the graph is the shared *density* node. The density node feeds both the Young’s modulus methods and the Srivastava Poisson’s ratio method through the same merge node. Because it is a single shared node, any pathway that uses the Srivastava method for Poisson’s ratio must use the same density method as is used for Young’s modulus — there is no independent density choice for Poisson’s ratio. This coupling, together with the four available density methods, four Young’s modulus methods, and two Poisson’s ratio methods, constrains D11 to $4 \times 4 \times 2 = 32$ unique calculation pathways. The complete graph structure is illustrated in Fig. ??.

80 2.4.2 Finding calculation pathways

All valid calculation pathways from the snowpit root to a target parameter are enumerated by a backward traversal algorithm. Beginning at the target parameter node, the algorithm recurses toward the root, applying the logic appropriate to each node type. At parameter nodes (OR logic), each incoming method edge is explored independently; each edge represents an alternative pathway for computing the parameter. At merge nodes (AND logic), sub-pathways from all required input branches are combined using a Cartesian product, so every combination of upstream method choices is represented in the output. Results for each node are memoized after their first computation, preventing redundant traversal of shared subgraphs when the same node appears in multiple branches of the DAG.

Following traversal, the resulting path trees are converted to a structured list of pathway objects and deduplicated. The deduplication step identifies structurally distinct traversals that resolve to identical sets of (parameter, method) assignments — an artefact of the shared density node, which the Cartesian-product expansion may enumerate more than once — and retains only the first occurrence of each unique method fingerprint. Applied to D11, the enumeration returns 32 unique pathways.

2.4.3 Executing parameterizations

For each enumerated pathway, the execution engine applies a two-phase procedure to a `Slab` object — an ordered list of `Layer` objects carrying the measured field observations.

95 In the first phase (layer-level), the engine iterates over all layers in the slab and computes, in dependency order, the density, Young’s modulus, Poisson’s ratio, and shear modulus of each layer using the methods specified by the current pathway. Density values are cached per layer, keyed by layer index and method name, and reused across pathways that share the same density method, avoiding redundant computation. Parameters that depend on density (Young’s modulus, Poisson’s ratio via the Srivastava method, and shear modulus) are always computed afresh for each pathway, because their values — including the propagated uncertainty budget — differ between pathways that use different upstream density methods; caching them would
100 silently collapse distinct uncertainty budgets into a single incorrect result.

In the second phase (slab-level), the engine assembles the per-layer elastic properties computed in the first phase and applies the classical laminate theory formulae (Sect. 2.3) to compute A11, B11, D11, and A55 for the slab as a whole.

A pathway fails silently for a given slab if any required field observation is absent, or if any method’s validity conditions are not satisfied (for example, a grain form outside the method’s applicable range, or a density value outside its regression validity interval). Failures are recorded with the reason, enabling post-hoc diagnosis of coverage losses. For successful pathway–slab combinations, the method used to compute each parameter on each layer is recorded for provenance tracing.

2.5 Constructing slabs from ECT observations

To demonstrate the framework at population scale, a slab definition is required for each observation. Extended Column Tests with propagating fracture (ECTP) provide a natural slab definition: the recorded failure depth identifies the weak layer, and all snow above it constitutes a candidate slab. The ECTP subset is also the largest category of SnowPilot stability tests that includes a well-defined failure depth (Table 1). We emphasise that no claim is made about the relationship between estimated D11 and ECT outcome; ECTP results serve here solely as a large, community-sourced source of field-observed slab stratigraphies with identifiable weak layers.

115 SnowPilot CAAML XML files were parsed into `SnowPit` objects using the `SnowPylot` library (Sect. 2.1). For each ECTP result in a pit, the weak layer was identified by matching the recorded failure depth to the depth boundaries of each stratigraphic layer: the weak layer is the layer whose upper boundary is at or shallower than the failure depth and whose lower boundary lies strictly below it. The slab was then defined as all layers above the weak layer (excluding the weak layer itself), and a `Slab` object was constructed from those layers, carrying the measured hand hardness, grain form, grain size, and thickness of each layer, and the pit slope angle. A single pit contributes more than one slab if it contains multiple ECTP results at different depths.

This procedure yielded 14,951 ECTP slab stratigraphies. ECTP results for which no stratigraphic layers were identified above the failure depth were excluded. These 14,951 slabs form the population-scale dataset for the coverage and parameterization analyses presented in Sect. ??.

3.1 Coverage and data loss by parameterization

Even under the most favourable combination of parameterization methods, valid D11 estimates can be obtained for only 5.1 % of the 14,951 ECTP slab stratigraphies in the dataset. This coverage ceiling is the primary finding of the analysis and sets an upper bound on the practical reach of field-data-driven mechanical stability models applied to standard snowpit observations.

130 The low ceiling arises from a compounding of validity failures across each step of the calculation chain. At the layer level, individual methods cover substantial fractions of the 377,729 layers: hand-hardness-based density methods apply to 28–63 % of layers (Sect. ??), E-mod methods to 5–55 % depending on grain form acceptance and density range (Sect. ??), and Poisson’s ratio methods to 28–42 % (Sect. ??). Slab-level coverage is systematically lower because a slab succeeds only if every one of its constituent layers passes all validity checks simultaneously for a given pathway. This multiplicative requirement transforms
135 modest layer-level failure rates into severe slab-level failure rates. The measured-density (`data_flow`) pathway illustrates the compounding starkly: 2.8 % of layers carry a density observation coinciding with a layer boundary, yet slab-level density coverage reaches only 0.7 % (109 slabs) because density must be present for every layer in the slab simultaneously. By the time the D11 calculation chain is complete, `data_flow` pathways succeed for at most 17 slabs (0.1 %), as the requirement for a complete per-layer density measurement stack is nearly never met in practice. The complete coverage landscape across
140 all density, E-mod, ν , and D11 pathway combinations is shown in Fig. ?? . Per-parameter detail—coverage fractions, method-to-method differences in applicable grain forms and density ranges, and the within-method spread in estimated values—is presented in Sects. ??–??.

3.2 Density estimates across methods

At the layer level, the three empirical density methods differ in both coverage and the distribution of estimates (Fig. ??). ? Table 2 achieved the broadest reach, yielding density estimates for 239,080 of 377,729 layers (63.3 %), with a mean of 208 kg m^{-3} and a mean relative uncertainty of 16.4 %. The ? parameterization covered 203,677 layers (53.9 %), with a lower mean of 193 kg m^{-3} and a relative uncertainty of 17.1 %. ? Table 5 was the most restrictive, applying to 106,370 layers (28.2 %), with a mean of 199 kg m^{-3} and the highest relative uncertainty at 18.4 %. The measured-density pathway (`data_flow`) was applicable to only 10,692 layers (2.8 %)—those with a measured density observation coinciding with a layer boundary—but
150 yielded a substantially higher mean of 283 kg m^{-3} and the lowest relative uncertainty of 10.0 %, reflecting that the remaining uncertainty arises from the $\pm 10 \%$ density measurement protocol rather than from regression scatter. The higher mean density of the `data_flow` subset likely reflects a sampling bias toward denser, harder layers where snow scientists chose to record density observations.

At the slab level, complete density coverage—meaning that every layer above the ECTP failure depth can be assigned a density estimate—is achievable for a smaller fraction of slabs than layer-level coverage alone would suggest, because a slab succeeds only if all of its constituent layers pass the grain form and hardness validity checks simultaneously. Of 14,951 ECTP slabs, ? Table 2 yields complete slab density estimates for 6,030 slabs (40.3 %), followed by ? at 4,598 slabs (30.8 %), ? Table 5 at 1,170 slabs (7.8 %), and the measured-density pathway at 109 slabs (0.7 %). The ordering of methods by slab-level coverage mirrors their layer-level ordering, suggesting that coverage loss at the slab level is driven primarily by the same grain
160 form restrictions that limit layer-level applicability, compounded by the requirement that no layer in a multi-layer slab may be excluded.

The $\sim 90 \text{ kg m}^{-3}$ spread in mean density across methods ($193\text{--}283 \text{ kg m}^{-3}$) is consequential for downstream calculations, because all implemented Young’s modulus parameterizations take density as their primary input. Even before any E-mod method uncertainty is considered, this method-to-method density disagreement introduces a systematic source of inter-pathway
165 variance. Section ?? quantifies how this density spread propagates into Young’s modulus estimates.

3.3 Young’s modulus estimates across methods

The 16 calculation pathways to Young’s modulus arise from combining 4 density sub-pathways with 4 E-mod methods (Fig. ??). Each E-mod method imposes its own grain form validity requirements on top of the grain form restrictions already

applied at the density step, so layer-level E-mod coverage is bounded above by the density coverage of the corresponding sub-pathway and generally falls below it. ? accepts the broadest set of grain form classes (DF, RG, FC, DH, MF) and a density range of 103–544 kg m⁻³, resulting in the highest layer-level coverage among all E-mod methods: the ? Table 2 sub-pathway yields 208,331 of 377,729 layers (55.2 %). Its power-law exponent of $n = 2.34$ is the lowest among the four methods, so density uncertainty is amplified by a relatively modest factor; within-pathway relative uncertainties for ? pathways are 33.8–38.5 % for hand-hardness density sub-pathways and 23.4 % for the measured-density sub-pathway. ? also achieves broad layer-level coverage (48.8 %) by accepting DF, RG, FC, DH, and SH grain forms, but its grain-form-specific coefficient pairs (A , n) carry large standard errors ($\sigma_A/A \approx 40$ –75 %; $\sigma_n/n \approx 12$ –24 %), and together with a high exponent ($n = 4.6$ –5.1), produce within-pathway uncertainties of 77–91 %. ? accepts RG, DH, and MF grain forms over 150–450 kg m⁻³ and covers 19.0–27.5 % of layers for hand-hardness sub-pathways; its purely exponential form produces high sensitivity to density errors and within-pathway uncertainties of 67–92 %. ? is the most restrictive, accepting only PP, RG, and DF grain forms within 110–363 kg m⁻³, covering 23.7–23.9 % of layers for the two highest-coverage hand-hardness sub-pathways, with within-pathway uncertainties of 72–77 %.

The inter-method spread in layer-level mean E is large (Fig. ??). For hand-hardness density sub-pathways, ? produces the highest mean values (281–311 MPa), followed by ? (71–134 MPa), with ? and ? yielding substantially lower means of 12–19 MPa and 7–13 MPa, respectively. The measured-density sub-pathway yields higher means across all E-mod methods (wautier 656 MPa, kochle 250 MPa, schottner 83 MPa, bergfeld 27 MPa), consistent with the elevated mean density of measured-density layers (Sect. ??).

At the slab level, complete E-mod coverage requires every layer in a slab to obtain a valid E-mod estimate in addition to a valid density estimate. The highest slab-level coverage is achieved by ? pathways: both the ? → schottner and ? Table 2 → schottner pathways succeed for 2,814 of 14,951 ECTP slabs (18.8 %). The ? Table 2 → ? pathway follows at 2,134 slabs (14.3 %), while ? pathways cover only 41–530 slabs (0.3–3.5 %) and measured-density sub-pathways fewer than 60 slabs (0.1–0.4 %). The full 16-pathway coverage matrix is presented in Fig. ??.

The roughly 15–40-fold spread in mean layer-level E between ? (~300 MPa) and ?/? (~8–15 MPa) for hand-hardness density sub-pathways is the dominant source of variance in D11 across the 32 pathways. Because D11 is linear in the layer-by-layer E values through the CLT integration, this inter-method E factor propagates directly as the same multiplicative factor in D11. Poisson’s ratio, examined in Sect. ??, varies less across methods and contributes secondarily to D11 spread (Sect. ??).

3.4 Poisson’s ratio estimates across methods

Five calculation pathways to Poisson’s ratio arise from combining two methods with their respective density sub-pathway requirements. The ? method requires no density input: it assigns a grain-form-specific mean ν value (RG: 0.171 ± 0.026 ; FC: 0.130 ± 0.040 ; DH: 0.087 ± 0.063) derived from the same FE simulations of m-CT snow microstructure used for the kochle E-mod method. Because density is not involved, the kochle ν pathway generates a single branch and achieves the broadest layer-level coverage: 159,120 of 377,729 layers (42.1 %), with an overall mean ν of 0.144 (reflecting the grain-form composition of the RG-/FC-/DH-applicable subset). The ? method also uses a grain-form-specific lookup (RG: 0.191 ± 0.008 ; PP and DF: 0.132 ± 0.053 ; FC and DH: 0.170 ± 0.020), but accepts five grain form classes and imposes a density validity threshold ($\rho > 200$ kg m⁻³). Because the density threshold is enforced by the graph, the srivastava pathway requires a density sub-pathway, producing four branches—one per density estimation method. Layer-level coverage for the srivastava hand-hardness sub-pathways is 28.0–29.4 % (105,757–110,937 layers), reflecting both grain form and density-threshold restrictions; the ? Table 5 and measured-density sub-pathways cover 10.8 % and 1.5 %, respectively. Mean ν across srivastava sub-pathways is 0.170–0.178. A notable property of both methods is that propagated uncertainty (with method uncertainty excluded) is exactly zero: because ν is assigned as a fixed value per grain form category rather than computed as a function of continuous density, the density measurement uncertainty does not propagate into ν .

At the slab level, kochle ν succeeds for 944 of 14,951 ECTP slabs (6.3 %).

The structural coupling introduced by the shared density node (Sect. ??) has a direct consequence for D11 pathway enumeration. When srivastava ν is selected, the density sub-pathway is already determined by the E-mod step (both share the same merge_density_grain_form node in the DAG), so no additional density branching occurs. When kochle ν is selected, ν is independent of the density sub-pathway entirely. In either case, the ν method choice adds exactly one branch to each of the

16 E-mod pathways, yielding $16 \times 2 = 32$ unique D11 pathways in total. This means that srivastava does not multiply the pathway count by 4 (one per density sub-pathway) at the D11 level—the coupling collapses what would otherwise be additional branching into a single constrained choice per E-mod pathway.

3.5 Bending stiffness (D11) of ECT slabs across all pathways

220 For the 14,951 ECTP slab stratigraphies assembled from the SnowPilot database, valid D11 estimates can be obtained for at most 5.1 % of slabs under any implemented pathway (Fig. ??). The maximum of 756 slabs is achieved by four pathways sharing a common structural feature: the kochle Poisson’s ratio method (which requires only grain form, not a density estimate), combined with either the ? or ? E-mod method (both accepting five grain form classes) and either the ? or ? Table 2 density sub-pathway (the two broadest hand-hardness methods). These combinations maximise coverage by avoiding the grain size requirement of ? Table 5, the density measurement requirement of the `data_flow` sub-pathway, and the grain form restrictions of the ? and ? E-mod methods. Pathways that require a direct density observation (`data_flow`) succeed for at most 17 slabs (0.1 %), because a layer-by-layer density measurement must be present for every layer in a slab simultaneously, and such complete density coverage within a single slab is rare in the SnowPilot database. The remaining 94.9 % of ECTP slabs—spanning more than a decade of observations from avalanche professionals and snow scientists—yield no valid D11 estimate under any of the 32 implemented pathways. This is not a limitation of the framework: the framework successfully identifies valid pathways wherever one exists. It is a gap in the observations: standard snowpit recording protocols do not routinely capture the combination of grain form, hand hardness, layer thickness, and Poisson’s-ratio-compatible grain types that D11 estimation requires for every layer.

For the slabs where D11 can be computed, the result depends strongly on which parameterization pathway is used (Fig. ??).
 235 Considering the four highest-coverage pathways (756 slabs each, all using kochle ν), mean slab D11 ranges from 8.7×10^7 N m (? Table 2 $\rightarrow$? $\rightarrow$ kochle) to 2.3×10^9 N m (? $\rightarrow$? $\rightarrow$ kochle)—a factor of ~ 26 arising solely from the choice of E-mod method applied to the same set of field observations. Across all 32 pathways, mean D11 spans from $\sim 3.6 \times 10^7$ N m (`data_flow` $\rightarrow$? $\rightarrow$ kochle, 6 slabs) to $\sim 5.0 \times 10^9$ N m (`data_flow` $\rightarrow$? $\rightarrow$ srivastava, 12 slabs). Isolating the contribution of each method choice: fixing density and ν method, the E-mod method contributes a 26-fold spread (wautier vs. schottner or bergfeld); fixing density and E-mod method, the ν method (kochle vs. srivastava) contributes a factor of ~ 1.4 ; and fixing E-mod and ν method, density method choice contributes a factor of ~ 1.0 – 1.1 . E-mod parameterization choice is thus the dominant source of inter-pathway D11 variance, with ν and density method choices playing secondary roles.

Within any single pathway, D11 estimates also carry substantial uncertainty attributable to propagated field measurement imprecision. Within-pathway relative uncertainties (input measurement propagation only, excluding method regression uncertainty) range from 23 % (`data_flow` $\rightarrow$? $\rightarrow$ kochle, 17 slabs) to 97 % (geldsetzer $\rightarrow$ kochle $\rightarrow$ srivastava, 344 slabs); for the four highest-coverage pathways, within-pathway uncertainties are 37–75 %. At $\pm 1 \sigma$, this within-pathway uncertainty represents a multiplicative factor of 1.4–2.7 on a single D11 estimate, substantially smaller than the 26-fold inter-pathway spread but not negligible. The inter-pathway spread (26-fold) and the within-pathway spread ($\pm 1 \sigma$ factor 1.4–2.7) are therefore distinct and complementary components of the total D11 uncertainty budget: the former reflects disagreement among published parameterizations about the physical relationship between field observables and mechanical properties, while the latter reflects the limited precision of those field observables themselves. Any D11 estimate derived from field observations carries both sources simultaneously, and neither can be reported without the other if the estimate is to be used as input to a quantitative mechanical stability model.

4 Discussion

255 4.1 Implications for field-data-driven mechanical stability models

4.2 Toward a preferred calculation pathway

4.3 Should mechanical properties be measured directly in the snowpit?

4.4 Limitations and assumptions

4.5 Extensions and future work

260 5 Conclusions

Code availability. SnowPyt-MechParams is openly available at [URL TBD]. SnowPylot is openly available at [URL TBD].

Data availability. The SnowPilot database is publicly accessible at snowpilot.org. The CAAML export files used in this analysis cover the period 1 September 2014 to 31 August 2025.

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